

FORM 1 THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT <i>(See section 7, 54 and 135 and sub-rule (1) of rule 20)</i>				(FOR OFFICE USE ONLY)	
			Application No.		
			Filing date:		
			Amount of Fee Paid:		
			CBR No:		
			Signature:		
1. APPLICANT'S REFERENCE / IDENTIFICATION No. (As Allotted By Office)					
2. TYPE OF APPLICATION [Please tick (✓) at the appropriate category]					
Ordinary (✓)		Convention ()		PCT-NP ()	
Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()
3A. APPLICANT(S)					
Name in Full		Nationality	Country of Residence	Address of the Applicant	
JEPPIAAR INSTITUTE OF TECHNOLOGY (JIT)		Indian	India	House No.	Principal, Jeppiaar Institute of Technology (JIT)
				Street	Kunnam, Sunguvarchatram, Kanchipuram District
				City	Sriperumbudur
				State	Tamil Nadu
				Country	India
				Pin code	631 604
3B. CATEGORY OF APPLICANT [Please tick (✓) at the appropriate category]					
Natural Person(✓)		Other than Natural Person			
		Small Entity ()		Startup ()	Others ()
4. INVENTOR(S) [Please tick (✓) at the appropriate category]					
Are all the inventor(s) same as the applicant(s) named above?			Yes()		No (✓)
If "No", furnish the details of the inventor(s)					
Name in Full		Nationality	Country of Residence	Address of the Applicant	
Dr. R. KALAIARASAN S. SANJAY A. SRI RAGAV KUMAR K. NIVETHIKA T. NANDHINI M. RAGHAVI		Indian	India	House No.	Dept of IT, Jeppiaar Institute of Technology
				Street	Kunnam, Sunguvarchatram,
				City	Sriperumbudur
				State	Tamil Nadu
				Country	India
				Pin code	631 604
				5. TITLE OF THE INVENTION	
Novel System, Design and Method of AI Bug Hunter: Intelligent Bug Detection and Security Analysis					
6. AUTHORISED REGISTERED PATENT AGENT(S)			IN/PA No.		
			Name		
			Mobile No.		

7. ADDRESS FOR SERVICE OF APPLICANT IN INDIA		Name	Dr. J. VENU GOPALA KRISHNAN					
		Postal Address	Principal, Jeppiaar Institute of Technology (JIT), Kunnam, Sunguvarchatram, Sriperumbudur, Tamil Nadu, India-631 604					
		Telephone No.	+91- 044-27159000					
		Mobile No.	74012 22007					
		Fax No.	+91- 044-27159006					
		E-mail ID	principal@jeppiaarinstitute.org					
8. IN CASE OF APPLICATION CLAIMING PRIORITY OF APPLICATION FILED IN CONVENTION COUNTRY, PARTICULARS OF CONVENTION APPLICATION								
Country	Application Number	Filing Date	Name of the Applicant	Title of the invention	IPC (as classified in the Convention Country)			
9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)								
International Application Number			International Filing Date					
10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION								
Original (first) application No.			Date of filing of original (first) application					
11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT								
Main application/patent No.			Date of filing of main application					
12. DECLARATIONS								
(i) Declaration by theinventor(s) <i>(In case the applicant is an assignee:</i> the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period). I/We, the above named inventor(s) is/are the true & first inventor(s) for this Invention and declare that the applicant(s) herein is/are my/our assignee or legal representative. Dated this 09th July 2026 Dr. R. KALAIARASAN S. SANJAY A. SRI RAGAV KUMAR K. NIVETHIKA T. NANDHINI M. RAGHAVI								
(ii) Declaration by the applicant(s) in the convention country (In case the applicant in India is different than the applicant in the convention country:the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period) I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative. Dated this 09th July 2026 Dr. J. VENU GOPALA KRISHNAN								

(iii) Declaration by the applicant(s)

I/We the applicant(s) hereby declare(s) that: -

- ✓ I am/ We are in possession of the above-mentioned invention.
- ✓ The provisional/complete specification relating to the invention is filed with this application.
 - ☐ The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by me/us before the grant of patent to me/us.
- ✓ There is no lawful ground of objection(s) to the grant of the Patent to me/us.
- ✓ I am/we are the true & first inventor(s).
- ✓ I am/we are the assignee or legal representative of true & first inventor(s).
 - ☐ The application or each of the applications, particulars of which are given in Paragraph-8, was the first application in convention country/countries in respect of my/our invention(s).
 - ☐ I/We claim the priority from the above mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from which I/We derive the title.
 - ☐ My/our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph-9.
 - ☐ The application is divided out of my /our application particulars of which is given in Paragraph-10 and pray that this application may be treated as deemed to have been filed on DD/MM/YYYY under section 16 of the Act.
 - ☐ The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph-11.

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
Complete/Provisional specification)#	5		
No. of Claim(s) and No. of Pages	7 & 1		
Abstract	1		
No. of Drawing(s) and No. of Pages	4 & 5		

In case of a complete specification, if the applicant desires to adopt the drawings filed with his provisional specification as the drawings or part of the drawings for the complete specification under rule 13(4), the number of such pages filed with the provisional specification are required to be mentioned here.

(b) Complete specification (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).

(c) Sequence listing in electronic form

- (d) Drawings (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).
- (e) Priority document(s) or a request to retrieve the priority document(s) from DAS (Digital Access Service) if the applicant had already requested the office of first filing to make the priority document(s) available to DAS.
- (f) Translation of priority document/Specification/International Search Report/International Preliminary Report on Patentability.
- (g) Statement and Undertaking on **Form 3**
- (h) Declaration of Inventorship on **Form 5**
- (i) Power of Authority
- (j).....

Total Fee Rs.....in Cash/ Banker's Cheque /Bank Draft bearing No.....

Date.....on BANK.

I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters slated herein are correct and I/We request that a patent may be granted to me/us for the said invention.

Dated this 09th July 2026

Dr. J. VENU GOPALA KRISHNAN

**To,
The Controller of Patents
The Patent Office, at Chennai**

Note:

- * Repeat boxes in case of more than one entry.
- * To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned.
- * Tick (✓) /cross (x) whichever is applicable/not applicable in declaration in paragraph-12.
- * Name of the inventor and applicant should be given in full, family name in the beginning.
- * Strike out the portion which is/are not applicable.
- * For fee: See First Schedule";

FORM 2
THE PATENTS ACT, 1970
(39 of 1970)
&
COMPLETE SPECIFICATION
(See section 10 and rule 13)

1. TITLE OF THE INVENTION Novel System, Design and Method of AI Bug Hunter: Intelligent Bug Detection and Security Analysis		
2. APPLICANT		
a) Name	b) Nationality	c) Address
JEPPIAAR INSTITUTE OF TECHNOLOGY (JIT)	Indian	Principal, JEPPIAAR INSTITUTE OF TECHNOLOGY (JIT), Kunnam, Sunguvarchatram, Sriperumbudur-631 604, Tamil Nadu, INDIA
3. PREAMBLE TO THE DESCRIPTION		
COMPLETE		
The following specification particularly describes the invention and the manner in which it is to be performed		
4. DESCRIPTION (Description shall start from next page)		
5. CLAIMS (Claims should start with the preamble – “I/We claim” on separate page)		
6. DATE AND SIGNATURE (to be given on the last page of specification)		
7. ABSTRACT OF THE INVENTION (to be given along with complete specification on the separate page)		
Note: *Repeat boxes in case of more than one entry *To be signed by the applicant(s) or the authorized registered patent agent *Name of the applicant should be given in full, family name in the beginning *Complete address of the applicant should be given stating with postal index no. / code, state and country *Strike out the column which is/are not applicable		

4. DESCRIPTION

This patent disclosure covers Novel System, Design and Method of AI Bug Hunter: Intelligent Bug Detection and Security Analysis.

AI Bug Hunter is a security tool that helps developers find problems in their code. It looks for vulnerabilities, secrets that are out in the old code that needs to be updated. This happens before the software is even deployed.

USERS CAN UPLOAD THEIR CODE THROUGH A WEBSITE:

1. Users can easily upload source code through a simple and user-friendly web interface.
2. The platform supports multiple upload formats, including project folders, ZIP files, and Git repositories.
3. Uploaded files are automatically checked and prepared for security scanning.
4. The system securely stores project information before starting the analysis process.
5. All uploaded code is analyzed within the platform, helping protect source code privacy while providing accurate security results.

AI Bug Hunter Workflow



Fig 1 Flow chart

AI BUG HUNTER: INTELLIGENT BUG DETECTION AND SECURITY ANALYSIS PLATFORM

SYSTEM ARCHITECTURE DIAGRAM

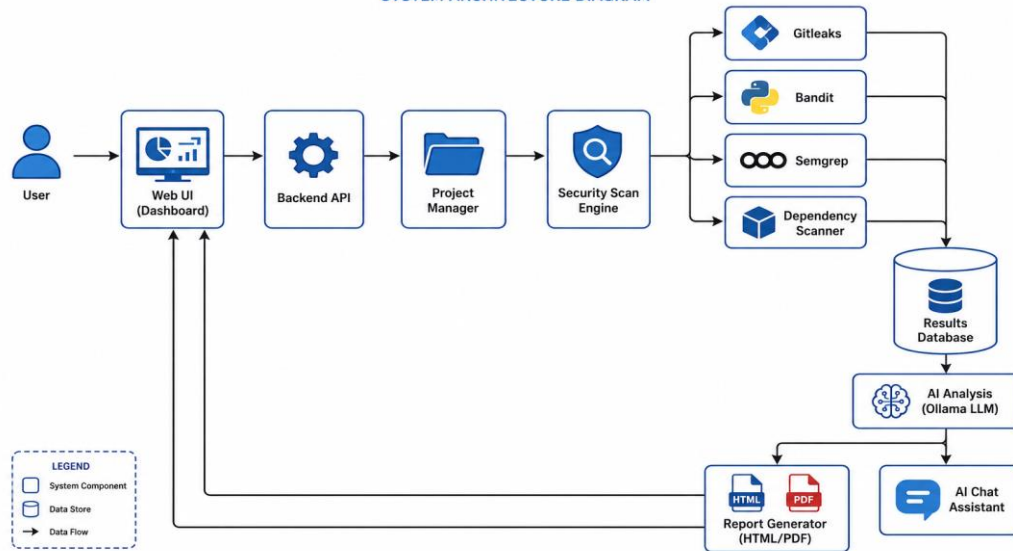


Fig 2 Architectural Diagram

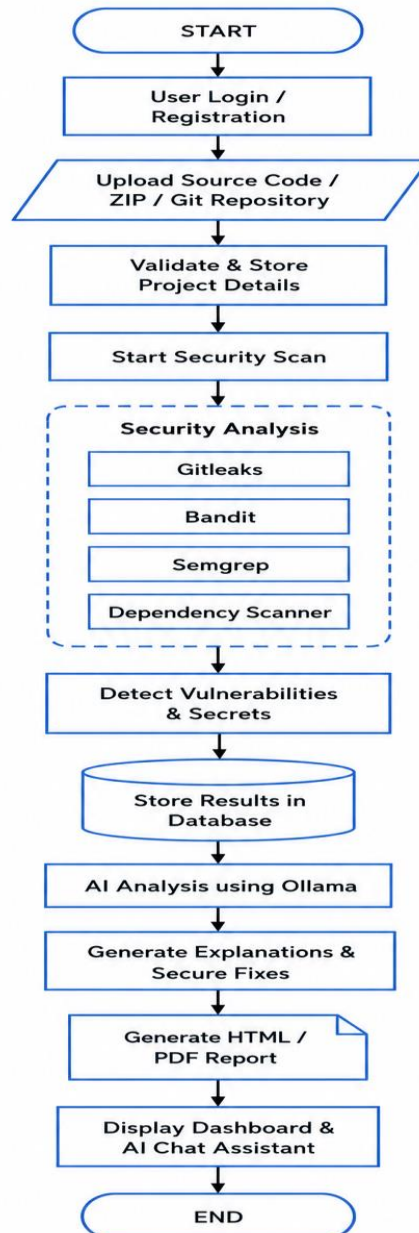


Fig 3 Flow Diagram

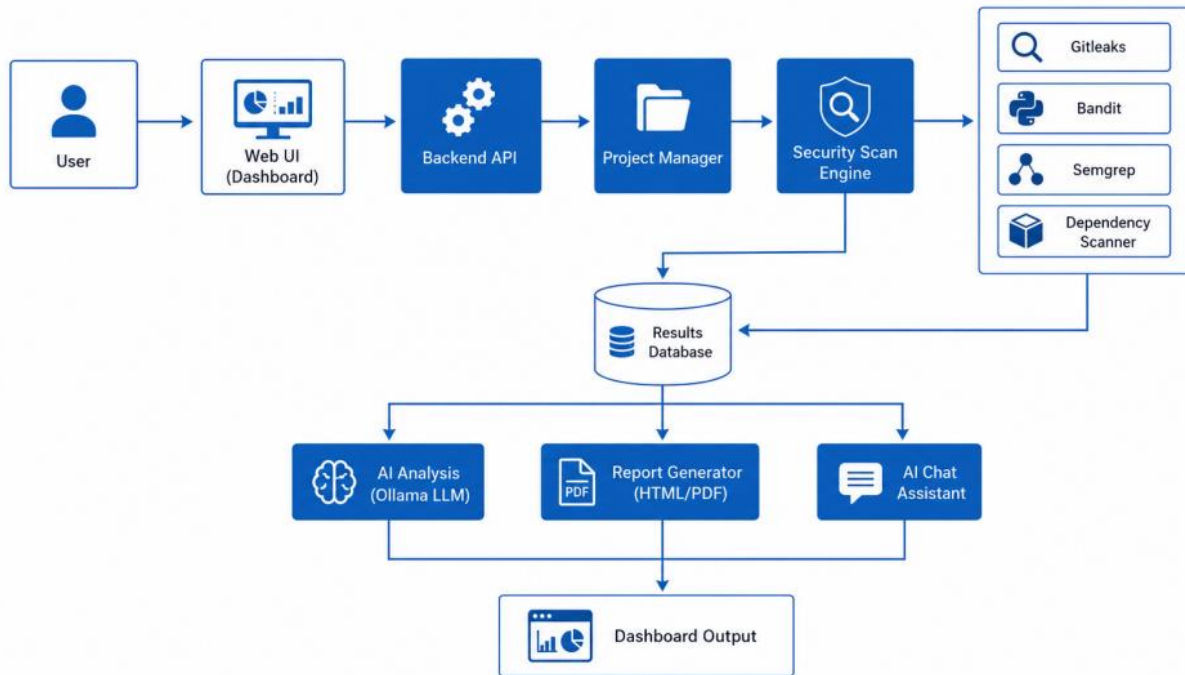


Fig 4 Block Diagram

Innovations

- **AI-Driven Security Analysis:** Integrates a locally hosted Large Language Model (LLM) to intelligently explain software vulnerabilities and recommend secure coding practices.
- **Privacy-Preserving Code Scanning:** Performs all security analysis locally using Ollama, ensuring that sensitive source code never leaves the organization's environment.
- **Unified Multi-Tool Security Framework:** Combines Gitleaks, Bandit, Semgrep, and dependency analysis into a single platform for comprehensive static application security testing.
- **Interactive AI-Assisted Vulnerability Remediation:** Provides developers with real-time explanations, risk assessments, and actionable remediation guidance through an intelligent AI assistant.
- **Automated Secure Development Workflow:** Supports multiple project upload formats, real-time scan monitoring, and automatic generation of detailed HTML/PDF security reports, streamlining the secure software development lifecycle.
- **Scalable Multi-Language Security Platform:** Enables early vulnerability detection across diverse programming languages, improving software quality, developer productivity, and cybersecurity readiness for students, startups, and enterprises.

Advantages:

1. **Early Vulnerability Detection** – Identifies security issues during the development phase.
2. **AI-Powered Analysis** – Provides intelligent explanations and secure code recommendations.
3. **Privacy Protection** – Uses a local AI model (Ollama), so source code is not sent to external servers.
4. **Multi-Language Support** – Analyzes applications written in different programming languages.
5. **Comprehensive Security Scanning** – Integrates Gitleaks, Bandit, Semgrep, and dependency analysis.
6. **Automated Reporting** – Generates detailed HTML and PDF security reports.
7. **Improved Code Quality** – Helps developers write more secure and reliable code.

8. **Time and Cost Efficient** – Reduces manual code review effort and accelerates vulnerability detection.
9. **Interactive AI Assistant** – Explains vulnerabilities and suggests secure fixes in an easy-to-understand manner.
10. **Scalable and Easy to Use** – Suitable for students, startups, enterprises, and software development teams.

REFERENCES:

- 1) Johnson B, Song Y, Murphy-Hill E, Bowdidge R. Why Don't Software Developers Use Static Analysis Tools to Find Bugs? Proceedings of the 2013 International Conference on Software Engineering (ICSE). 2013 May; pp. 672–681. doi: 10.1109/ICSE.2013.6606613.
- 2) Pashchenko I, Dashevskiy S, Massacci F. Understanding the Software Composition Analysis (SCA) Tools: A Systematic Mapping Study. IEEE Access. 2020; 8:148705-148722. doi: 10.1109/ACCESS.2020.3016429.
- 3) Livshits B, Lam MS. Finding Security Vulnerabilities in Java Applications with Static Analysis. Proceedings of the 14th USENIX Security Symposium. 2005 Aug; pp. 271–286.
- 4) Open Worldwide Application Security Project (OWASP). OWASP Dependency-Check: Software Composition Analysis Tool for Identifying Vulnerable Components. OWASP Foundation. 2023. Available as an official OWASP security tool for dependency vulnerability detection.
- 5) Bandit Development Team. Bandit: Security Linter for Python Source Code. OpenStack Security Project. 2024. Bandit performs static security analysis of Python applications by identifying common coding vulnerabilities and insecure programming practices.
- 6) Semgrep Inc. Semgrep: Fast Static Analysis for Finding Bugs, Security Vulnerabilities, and Enforcing Coding Standards. Semgrep Documentation. 2024. Supports multi-language static application security testing through customizable rule-based analysis.
- 7) Gitleaks Development Team. Gitleaks: Detecting and Preventing Hardcoded Secrets in Git Repositories. Open Source Security Tool. 2024. Provides automated detection of API keys, passwords, tokens, and other sensitive credentials within source code and Git repositories.

5. CLAIMS

I/We Claim,

1. The patent disclosure covers Novel System, Design and Method of AI Bug Hunter: Intelligent Bug Detection and Security Analysis as described above in Fig 1 to 4.
2. **AI-Driven Security Analysis:** Integrates a locally hosted Large Language Model (LLM) to intelligently explain software vulnerabilities and recommend secure coding practices.
3. **Privacy-Preserving Code Scanning:** Performs all security analysis locally using Ollama, ensuring that sensitive source code never leaves the organization's environment.
4. **Unified Multi-Tool Security Framework:** Combines Gitleaks, Bandit, Semgrep, and dependency analysis into a single platform for comprehensive static application security testing.
5. **Interactive AI-Assisted Vulnerability Remediation:** Provides developers with real-time explanations, risk assessments, and actionable remediation guidance through an intelligent AI assistant.
6. **Automated Secure Development Workflow:** Supports multiple project upload formats, real-time scan monitoring, and automatic generation of detailed HTML/PDF security reports, streamlining the secure software development lifecycle.
7. **Scalable Multi-Language Security Platform:** Enables early vulnerability detection across diverse programming languages, improving software quality, developer productivity, and cybersecurity readiness for students, startups, and enterprises.

Dated this 09th July 2026

Dr. J. VENU GOPALA KRISHNAN

Dr. R. KALAIARASAN

S. SANJAY

A. SRI RAGAV KUMAR

K. NIVETHIKA

T. NANDHINI

M. RAGHAVI

6. ABSTRACT OF THE INVENTION

The patent disclosure covers Novel System, Design and Method of AI Bug Hunter: Intelligent Bug Detection and Security Analysis. With the increasing use of software in everyday applications, protecting source code from security threats has become essential. This AI Bug Hunter, Intelligent Bug Detection and Security Analysis, introduces an AI-powered Static Application Security Testing (SAST) platform that enables developers to identify software dependencies at an early stage of development. This integrates trusted security tools, including Gitleaks, Bandit, and Semgrep, with a locally hosted Large Language Model (LLM) through Ollama to provide clear vulnerability explanations and practical secure coding recommendations while ensuring source code privacy. It also features real-time scan monitoring, interactive AI assistance, and comprehensive HTML/PDF security reports. By automating vulnerability detection and remediation guidance, the patent reduces manual effort, improves code quality, enhances developer productivity, and supports the development of secure and reliable software systems.

FORM 3
THE PATENTS ACT, 1970 (39 of 1970) and
THE PATENTS RULES, 2003
STATEMENT AND UNDERTAKING UNDER SECTION 8
(See section 8; Rule 12)

1. Name of the applicant(s).	I/We JEPPIAAR INSTITUTE OF TECHNOLOGY hereby declare:				
2. Name, address and nationality of the joint applicant.	(i) that I/We have not made any application for the same/substantially the same invention outside India Not Applicable Or (ii) that I/We who have made this application No.....datedalone/ jointly with, made for the same/ substantially same invention, application(s) for patent in the other countries, the particulars of which are givenbelow:				
Name of the Country	Date of Application	Application No.	Status of the Application	Date of Publication	Date of Grant
Not Applicable					
3. Name and address of the assignee	(iii) that the rights in the application(s) has/have been assigned to JEPPIAAR INSTITUTE OF TECHNOLOGY (Indian) Principal, Jeppiaar Institute of Technology (JIT), Kunnam, Sunguvarchatram, Sriperumbudur-631 604, Tamil Nadu, INDIA And that I/We undertake that up to the date of grant of the patent by the Controller, I/We would keep him informed in writing the details regarding corresponding applications for patents filed outside India within six months from the date of filing of such application. Dated this 09th July 2026				
4. To be signed by the applicant or his authorized registered patent agent.	Signature:				
5. Name of the natural person who has signed.	Dr. J. VENU GOPALA KRISHNAN				
	To The Controller of Patents, The Patent Office, at Chennai				
<i>Note: Strike out whichever is not applicable</i>					

FORM 5
THE PATENTS ACT, 1970
(39 of 1970)

DECLARATION AS TO INVENTORSHIP
[See section 10(6) and rule 13(6)]

1. Name(s) of the Applicant(s)

I/We, **Dr. R. Kalaiarasan, S. Sanjay, A. Sri Ragav Kumar, K. Nivethika, T. Nandhini and M. Raghavi** of the invention disclosed in the complete specification filed in pursuance of my/our application numbered datedis/are:

2. APPLICANT		
a) Name	b) Nationality	c) Address
Dr. R. KALAIARASAN	Indian	Department of IT, Jeppiaar Institute of Technology (JIT), Kunnam, Sunguvarchatram, Sriperumbudur Tamil Nadu, India-631 604
S. SANJAY	Indian	
A. SRI RAGAV KUMAR	Indian	
K. NIVETHIKA	Indian	
T. NANDHINI	Indian	
M. RAGHAVI	Indian	

Dated this 09th July 2026

Dr. R. KALAIARASAN

S. SANJAY

A. SRI RAGAV KUMAR

K. NIVETHIKA

T. NANDHINI

M. RAGHAVI

3. DECLARATION to be given when the application in India is filed by the applicant(s) in the convention country:

Not Applicable

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s)

Dated thisday of2026

Signature: **Not Applicable**

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form)

Not Applicable

I/We assent to the invention referred to in the above declaration, being included in the complete specification filed in pursuance of the stated application.

Dated thisday of2026

Signature of the additional inventor(s): **Not Applicable**

To,
The Controller of Patent
The Patent Office, Chennai

FORM 9
THE PATENTS ACT, 1970
(39 of 1970)
&
The Patents Rules, 2003
REQUEST FOR PUBLICATION
[See section 11A (2); rule 24A]

1. Name, address and nationality of the applicant(s).

a) Name	b) Nationality	Address
JEPPIAAR INSTITUTE OF TECHNOLOGY (JIT)	Indian	Principal, Jeppiaar Institute of Technology (JIT), Kunnam, Sunguvarchatram-631 604, Sriperumbudur, Tamil Nadu, INDIA Mobile: 74012 22007; Email: principal@jeppiaarinstitute.org

I/We **JEPPIAAR INSTITUTE OF TECHNOLOGY (Indian)** hereby request for early publication of my / our Application filed on **09/07/2026** for the invention titled **Novel System, Design and Method of AI Bug Hunter: Intelligent Bug Detection and Security Analysis** under section 11A (2) of the Act.

2. To be signed by the applicant or by his authorized registered patent agent.

3. Name of the natural person who has signed.

Dr. J. VENU GOPALA KRISHNAN (Indian)

Dated this 09th July 2026

Dr. J. VENU GOPALA KRISHNAN

To,
The Controller of Patent
The Patent Office, Chennai

Note: - For fee: - See First Schedule.

FORM 28
THE PATENTS ACT, 1970
(39 of 1970)
AND
THE PATENTS RULES, 2003
Patents (Amendment) Rules, 2021
TO BE SUBMITTED BY A SMALL ENTITY /STARTUP/EDUCATIONAL INSTITUTION
[See rules 2 (fa), 2(fb), 2(ca) and 7]

1 Insert name, address and nationality.

a) Name	b) Nationality	Address
Dr. J. VENU GOPALA KRISHNAN	Indian	Principal, Jeppiaar Institute of Technology (JIT), Kunnam, Sunguvarchatram-631 604, Sriperumbudur, Tamil Nadu, INDIA Mobile: 74012 22007; Email: principal@jeppiaarinstitute.org

I/We **Dr. J. VENU GOPALA KRISHNAN (Indian)** applicant/patentee in respect of the Patent Application No **filed** or Patent No -nil- hereby declare that I/we am/are a small entity in accordance with rule 2(fa) or a start-up in accordance with rule 2(fb) or an educational institution in accordance with rule 2(ca) and submit the following document(s) as proof:

2 Documents to be submitted

~~I. For claiming the status of a small entity:~~

~~A. For an Indian applicant: Evidence of registration under the Micro, Small and Medium Enterprises Development Act, 2006 (27 of 2006).~~

~~B. In case of a foreign entity: Any other document.~~

~~II. For claiming the status of a start-up~~

~~A. For an Indian applicant: Any document as evidence of eligibility, as defined in rule 2(fb).~~

~~B. In case of a foreign entity: Any other document.~~

~~III. For claiming the status of an educational institution~~

~~A. For an Indian applicant: Any document as evidence of eligibility, as defined in rule 2(ca). Copy of UGC, AICTE, Govt of India, Approval Attached and Anna University, Govt of Tamilnadu Letter Attached.~~

~~B. In case of a foreign educational institution: Any other document.~~

3 To be signed by the applicant(s) /patentee(s)/ authorized registered patent agent.

The information provided herein is correct to the best of my/our knowledge and belief.

Dr. J. VENU GOPALA KRISHNAN (Indian)

Dated this 09th July 2026

Dr. J. VENU GOPALA KRISHNAN

Principal

Jeppiaar Institute of Technology

Designation and official seal, if any, of the person who has signed.

To,

The Controller of Patent

The Patent Office, Chennai